WEEK 7. AI/DATA WORKFLOW & RISK REGISTER

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Goal: show how information supports a business decision and how Company B governs the workflow. Use the Week 6 customer/marketing use case where possible.

A. Business decision and systems

Decision/use case; who benefits; what system(s) are involved (e.g., CRM, MIS, analytics):
Decision: for the Gulf pilot (one product line, one UAE channel), which launch ad versions go to which customer group and language, and how much stock to send to each shop/channel. This continues the Week 6 use case: launch messages for halal-seeking Gulf families that answer "Is it halal? Is it good?" [4].

Who benefits: Gulf families (clear messages in their own language, product on the shelf); the export manager and importer (right stock, less waste); the brand manager (faster ad drafts). **Systems:** TPS for orders, invoices and deliveries; a marketing data mart (the marketing slice of the data warehouse); AI used as a decision support system (DSS) for forecasts and ad drafts; MIS monthly reports [1].

Data vs information: what raw data enters, what useful information is produced?
Raw data in: order and delivery records from the TPS (by shop, date, SKU); loyalty sign-ups only where people agreed (language preference, emirate, contact); public review and competitor-price summaries with no personal data; brand facts and halal/label rules [2], [4].

Useful information out: a demand forecast per shop and channel; a short list of liked flavours and box sizes; ad options per group in Arabic, English and Kazakh; and an MIS statement a manager can act on, e.g. "waffle sales in one channel fell this month" [1] (photo essay, Visual 8).

B. Workflow map

Draw or insert your diagram here: DATA SOURCE → COLLECTION / ACCESS → PROCESSING / AI ANALYSIS → HUMAN VERIFICATION → BUSINESS DECISION → CUSTOMER ACTION → MONITORING

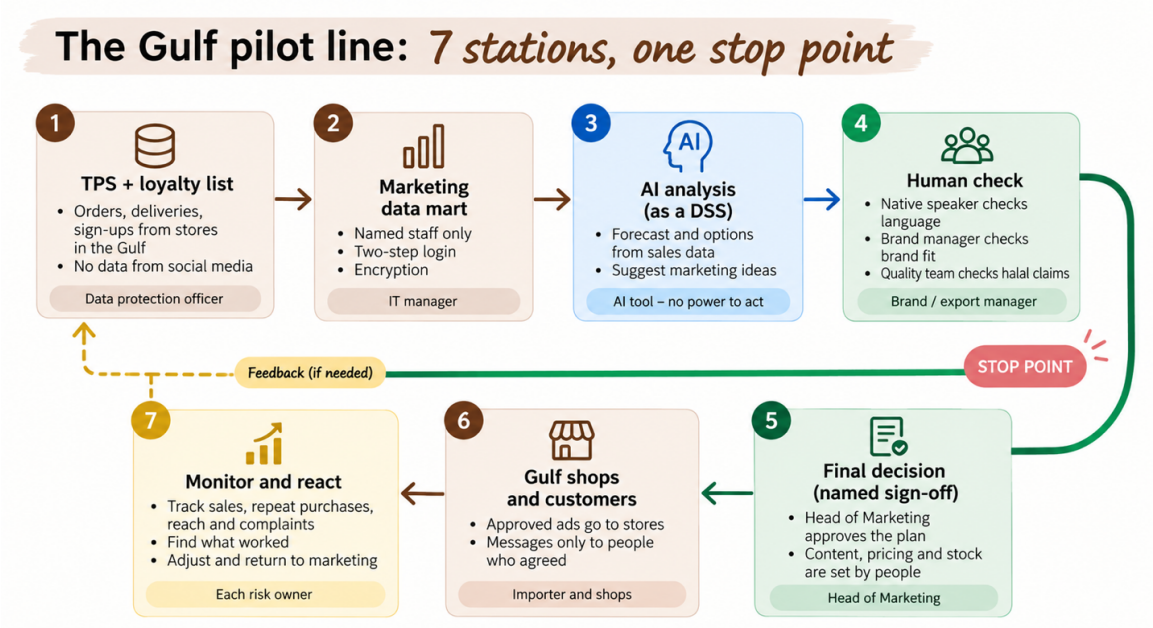


Figure: Gulf pilot line – 7 stations from data source to monitoring, with a stop point at the human check and a feedback line (photo essay, Visual 7; diagram drawn by the author in Canva).

DATA SOURCE (TPS orders, consented loyalty sign-ups) → **COLLECTION / ACCESS** (marketing data mart, named access) → **PROCESSING / AI ANALYSIS** (AI as DSS: forecast + ad drafts) → **HUMAN VERIFICATION** (native speaker, brand manager, export manager) → **BUSINESS DECISION** (Head of Marketing signs) → **CUSTOMER ACTION** (approved ad / stock reaches the UAE shop) → **MONITORING** (monthly checks, triggers, fallback).

For each stage: actor/system, inputs/outputs, permissions and data movement:

1 Data source – TPS + loyalty list. In: orders, deliveries, consented sign-ups. Out: raw records. Nothing is taken from social media [5].

2 Collection / access – marketing data mart (IT manager). Data moves from the TPS into the data mart; access by named staff only, two-step login, encryption [6].

3 Processing / AI analysis – AI tool as DSS. In: aggregated sales and brand facts – no names or contacts in prompts. Out: forecast and ad options. Permission: suggest only; it cannot publish, order or set prices.

4 Human verification – native speaker, brand manager, quality team, export manager. In: AI options. Out: corrected, approved options.

5 Business decision – Head of Marketing. Signs the final ad set and stock plan; every approval saved with name and date [5], [7].

6 Customer action – importer and UAE shops. Approved ads go out through the importer's channels; loyalty messages only to people who consented.

7 Monitoring – each risk owner. Monthly: forecast error, fair reach across groups and languages, complaints; results go back into planning.

Human review: who approves what, and using which criteria?

Native speaker + brand manager approve every ad: correct meaning, brand voice of a 50+ year brand, no health claims, no stereotypes. **Quality team** approves every halal/label claim: only claims backed by a valid certificate.

Export manager approves the stock plan: forecast compared with real distributor orders. **Head of Marketing** gives the final named sign-off; no name on a decision → the campaign pauses (photo essay, Visual 9).

Escalation for uncertain/high-risk cases; manual fallback for outages:

Escalation: uncertain or high-risk cases (a halal or health claim, a customer complaint, missing consent, a forecast far from real orders) go risk owner → Head of Marketing; AI never decides about a customer alone [5].

Manual fallback: if the AI tool or data mart is down, or forecast error is over 25% for 2 months, plan from distributor orders and an offline customer list, and use the Week 6 human-only route (agency + translator) for ads. Thresholds are the author's proposals, not company data.

C. Governance risk register — at least two substantive risks

Risk and affected data	Likelihood / impact	Control + owner	Monitoring / evidence
Privacy – customer data used without consent, or public posts reused for ads. Data: loyalty sign-ups (contact, language, emirate).	Likelihood: Medium. Impact: High – legal breach under 2026 rules [5] and loss of trust in a new market.	Use only consented sign-ups; never take data from social media; no personal data in AI prompts. **Owner: Data protection officer.**	Consent log; complaints register. Trigger: any complaint or missing consent → stop using that data, review within 48 h.
Reliability – wrong forecast, bad translation, or an ad with a false halal/health claim. Data: TPS sales, AI ad drafts.	Likelihood: High (new market, little sales history). Impact: High – wasted stock or a misleading claim.	Forecast compared with real orders; native speaker + brand manager approve every ad; quality team checks claims [7]. **Owner: Export manager.**	Monthly forecast-error report; approval log. Trigger: error > 25% for 2 months → plan from distributor orders.
Security – one data mart holds the whole	Likelihood: Low–Medium. Impact: High.	Encryption, named access, two-step login; breaches	Access log reviewed monthly; incident register;

customer list, so one leak exposes everyone [1]. Data: data mart, loyalty list.		reported under 2026 rules [6]. **Owner: IT manager.**	breach → notify and lock accounts.
Bias & accountability – some groups or languages get fewer offers; an AI output is used without anyone signing. Data: offer lists, ad sets.	Likelihood: Medium. Impact: Medium–High – unfair treatment; automated-only decisions are banned [5].	Monthly fairness check across groups and languages (Marketing manager); named sign-off on every decision. **Owner: Head of Marketing.**	Offers-per-group report; approval log with name and date. Trigger: no name on a decision → pause the campaign.

Consider privacy, security, bias, reliability, accountability, escalation and fallback. Do not list a risk without an actionable control.

D. Governance decision and transfer

Which risk matters most for Company B and why? What is the residual risk?
Privacy and consent matter most. Bayan Sulu is new in the Gulf and has little customer data; one misuse of personal data would damage trust before the brand is known, and Kazakhstan's 2026 amendments ban decisions made only by automated processing and tighten breach rules [5], [6].

Residual risk (medium-low): even with consent-only data and named access, the importer partner and staff still handle contact data, and AI drafts can still contain errors – so human review and monthly checks remain necessary.

Company A practice + verified source; transfer verdict and Kazakhstan-specific justification:
Ferrero practice: Ferrero built a consumer data platform on Google Cloud BigQuery ("Raw Data" by fifty-five) that splits its consumer database into real-time clusters for personalised content and media buying [8]. Ferrero USA also uses Alloy.ai for demand forecasting [10].

Verdict: ADAPT the BigQuery practice, PILOT forecasting (as in Week 6 and photo essay Visual 10). Kazakhstan-specific reasons: Bayan Sulu sells mainly through distributors and has far less online data than Ferrero; the 2026 personal-data amendments require consent-based processing, human involvement in decisions and breach reporting [5], [6]. So start with simple groups from distributor sales and consented sign-ups, with named human sign-off – not real-time automated personalisation.

Sources (IEEE; numbering follows the A3 photo essay and the Week 6 artifact):

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. Houston, TX, USA: OpenStax, 2026, chs. 11–13. [Online]. Available: https://openstax.org/details/books/introduction-business-2e

[2] JSC "Bayan Sulu," "About Us," bayansulu.kz. [Online]. Available: https://bayansulu.kz/en/about/o-kompanii/ (accessed Sep. 16, 2026).

[4] KazFoodProducts, "Bayan Sulu Confectionery Captivates Gulfood - 2026 and Expands into New Markets," press release, Jan. 30, 2026. [Online]. Available: https://kfp.kz/en/press-center/baian-sulu-konditerlik-fabrikasy-gulfood-2026-kormesin-bagyndyrdy (accessed Oct. 14, 2026).

[5] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026

[6] Baker McKenzie, "Kazakhstan: New Amendments to Personal Data Legislation," Jul. 21, 2026. [Online]. Available: https://www.bakermckenzie.com/en/insight/publications/2026/07/kazakhstan-new-amendments-to-personal-data-legislation

[7] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.

[8] Google Cloud, "Ferrero Case Study." [Online]. Available: https://cloud.google.com/customers/ferrero (accessed Sep. 17, 2026).

[10] Alloy.ai, “Ferrero USA Selects Alloy.ai for Digital Supply Chain Management.” [Online]. Available: https://alloy.ai/alloy-selected-by-ferrero-usa-for-digital-supply-chain-management (accessed Sep. 16, 2026).

Links to other work: Week 6 artifact (use case and comparison); photo essay Visuals 7 (workflow), 8 (data systems), 9 (risk controls) and 10 (transfer verdict).